# Key-Region Prototype Contrastive Learning for Few-Shot PAN–MS Land-Cover Classification

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***Abstract*—Few-shot panchromatic (PAN)–multispectral (MS) remote sensing image classification is challenging due to limited labeled samples and substantial cross-modal differences. PAN images provide rich spatial and structural details, whereas MS images contain discriminative spectral information. To address background interference and sample selection bias, we develop TK-Net to improve task construction and cross-modal representation learning under scarce supervision. TK-Net consists of two key components. First, a task-adaptive sampling (TAS) module replaces conventional random episodic sampling with a feature-driven strategy. It selects dispersed support samples and hard query samples near confusing decision boundaries, thereby improving prototype representativeness and boundary discrimination. Second, a key region mining and alignment (KMA) module identifies cross-modally consistent foreground regions through superpixel-guided region modeling and performs region-selective prototype contrastive alignment, thereby suppressing irrelevant background responses and improving cross-modal semantic consistency. Experiments on three dual-source remote sensing benchmarks demonstrate the effectiveness of TK-Net. Our code is available at: https://github.com/Xidian-AIGroup190726/TKnet**

***Index Terms*—Few-shot learning (FSL), panchromatic (PAN) images and multispectral (MS) images, prototype contrastive learning, task-adaptive sampling (TAS).**

## I. Introduction

SATELLITE remote sensing imagery has become an important data source for land-cover mapping and scene interpretation, supporting applications such as environmental monitoring, resource management, and urban analysis [1], [2]. Recent studies on remote sensing change captioning also indicate the growing demand for semantic-level visual understanding beyond pixel-wise recognition [3]. Among different data modalities, panchromatic (PAN) and multispectral (MS) images provide complementary information: PAN images capture fine spatial details, while MS images offer rich spectral cues for semantic discrimination. However, substantial differences in spatial resolution and imaging mechanisms between the two modalities make simple fusion strategies insufficient [4], [5], [6]. Additionally, pixel-level annotation in remote sensing is costly and labor-intensive, making large-scale labeled data difficult to obtain. This data scarcity poses severe challenges to reliable cross-modal classification under few-shot settings [7], [8].

These challenges are also relevant to disaster emergency response, where remote sensing systems must rapidly interpret newly acquired scenes with limited labeled data; recent studies have explored flood monitoring and spatiotemporal change detection [9], [10]. Although TK-Net is evaluated on few-shot PAN–MS land-cover classification rather than directly on flood or building change detection, its cross-modal fusion and region-aware representation learning may provide a useful foundation for downstream emergency-response tasks, where reliable boundaries and structurally informative regions are important.

Benefiting from the complementary characteristics of PAN and MS imagery, multimodal remote sensing scene classification has been extensively studied [11], [12]. Existing PAN–MS integration methods can be roughly grouped into data-level, feature-level, and decision-level strategies. Feature-level fusion integrates intermediate deep embeddings and has emerged as the most widely adopted paradigm [6], [11]. Recent multimodal studies have further explored modality-specific information preservation and guided fusion to improve the use of complementary spectral and spatial cues [13], [14]. However, these methodologies typically rely on abundant annotated data to optimize complex multimodal networks. When confronted with extremely limited training samples, they struggle to establish reliable cross-modal correspondences and are highly susceptible to severe overfitting, limiting their practical deployment in few-shot scenarios.

To address the scarcity of labeled data, recent studies have explored several training paradigms tailored for few-shot learning (FSL). Existing FSL solutions for remote sensing can be viewed from three main perspectives: knowledge transfer, metric learning, and task construction [7], [15].

Transfer learning has been widely adopted to mitigate the limited-sample bottleneck [16], [17]. In remote sensing, few-shot transfer has also been studied for SAR image classification [18]. Nevertheless, transferred global features may exhibit limited intraclass compactness and interclass separability, constraining generalization to novel categories.

Metric-based few-shot methods classify novel categories by learning discriminative embeddings and measuring similarities against class prototypes [19], [20], [21], [22]. In [23] and [24], contrastive metric learning and cross-source similarity constraints were introduced to enhance representation discriminability. Other works [25], [26] focus on alleviating cross-source misalignment via modality-specific feature modeling. Nevertheless, since these methods

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predominantly rely on global or sample-level feature representations, they may overlook fine-grained, region-aware semantic correspondences. Consequently, their alignment processes can become susceptible to task-irrelevant background noise, restricting robustness when dealing with complex cross-modal heterogeneities under extremely low-sample conditions.

Task-driven few-shot methods, often based on meta-learning, aim to train models capable of rapid adaptation via episodic training [15], [27], [28]. Works such as [29], [30], [31] explored dual-adjustment frameworks, adversarial task augmentation, and lightweight multiscale adaptation to enrich meta-training tasks and improve generalization. However, the task construction process in these frameworks typically relies on uniform random sampling, which treats all instances equally and may not fully account for sample-level discriminative difficulty or intraclass feature dispersion. Consequently, randomly generated episodes might lack the challenging boundary instances needed to effectively optimize the model, constraining discriminative capacity under extremely data-scarce conditions.

While the aforementioned studies have made significant progress, they still face inherent methodological bottlenecks in practical applications.

1) *Insufficient Boundary-Aware Task Construction:* Most FSL methods construct episodic tasks through uniform random sampling, where all samples are treated equally. However, remote sensing scenes often exhibit significant intraclass variation and subtle interclass differences. Randomly sampled episodes may be dominated by easily distinguishable samples, while boundary samples near confusing categories are not sufficiently explored. Consequently, supervision for class separation is weakened, reducing model generalization when labeled data are scarce.
2) *Coarse Cross-Modal Alignment Under Background Interference:* PAN and MS images exhibit inherent spectral–spatial discrepancies, and their local semantic responses are not always consistent. Existing methods usually reduce this discrepancy by aligning global features. However, global alignment inevitably mixes discriminative foreground regions with irrelevant background responses. Under few-shot supervision, such noisy aggregation can distort cross-modal correspondences and produce unreliable prototypes, making metric-based classification less robust.

Moreover, naively combining prototype and contrastive learning does not necessarily produce complementary benefits. Prototypes from uninformative episodes may provide unreliable contrastive anchors, while global contrastive constraints can amplify background interference. Therefore, effective integration requires coherent coupling between task construction, prototype quality, and the spatial scope of contrastive alignment.

Motivated by these limitations, we develop a few-shot cross-modal framework for remote sensing image classification. Our primary contributions are as follows.

1) *Task-Adaptive Sampling:* We propose a task-adaptive sampling (TAS) module that replaces random sampling with a feature-driven and difficulty-aware strategy. TAS constructs support sets with larger intraclass dispersion and query sets closer to decision boundaries, exposing the model to more informative episodes for stronger boundary discrimination and better generalization under limited supervision.
2) *Key Region Mining and Alignment:* We design a key region mining and alignment (KMA) module for fine-grained cross-modal alignment. KMA explicitly identifies informative foreground regions and suppresses background interference during cross-modal interaction. A region-selective prototype contrastive mechanism is introduced to generate cleaner prototypes and improve class separability, enhancing the robustness of metric-based classification in few-shot scenarios.

The remainder of this article is arranged as follows. Section II provides a review of related work. Section III introduces the proposed framework and its main technical components. Section IV presents the datasets, experimental settings, and results used to evaluate the effectiveness of our approach.

## II. Related Work

### A. FSL and Prototypical Networks

Prototypical networks [20] serve as a foundational metric-based paradigm in FSL. In this framework, class-level prototypes are constructed by aggregating the features of support set samples belonging to the same class, and query classification is performed based on their metric distances to these prototypes. By providing compact and representative semantic descriptions of classes, this method exhibits strong stability under limited sample conditions.

Building upon this paradigm, several studies have extended prototypical networks to few-shot remote sensing image classification. For example, SPNet [32] combines Siamese networks with prototype learning to achieve effective scene recognition under few-shot settings. Other works further enhance prototype representations by introducing dynamic prototype modeling or spatial convolution operations to capture complex structures better [33]. However, these methods primarily rely on global features for prototype construction, and their episodic training heavily depends on uniform random task sampling. Consequently, they often overlook the varying discriminative difficulty of samples, and rich structural information in the feature space is not fully exploited, inherently constraining their generalization capability in complex cross-modal remote sensing scenarios.

### B. Cross-Modal Alignment and Key Region Modeling in Remote Sensing

In multisource and cross-modal remote sensing tasks, severe distribution discrepancies across modalities render feature alignment a critical challenge. To bridge these semantic gaps, various feature-level alignment strategies have been explored. For instance, global–local interaction networks

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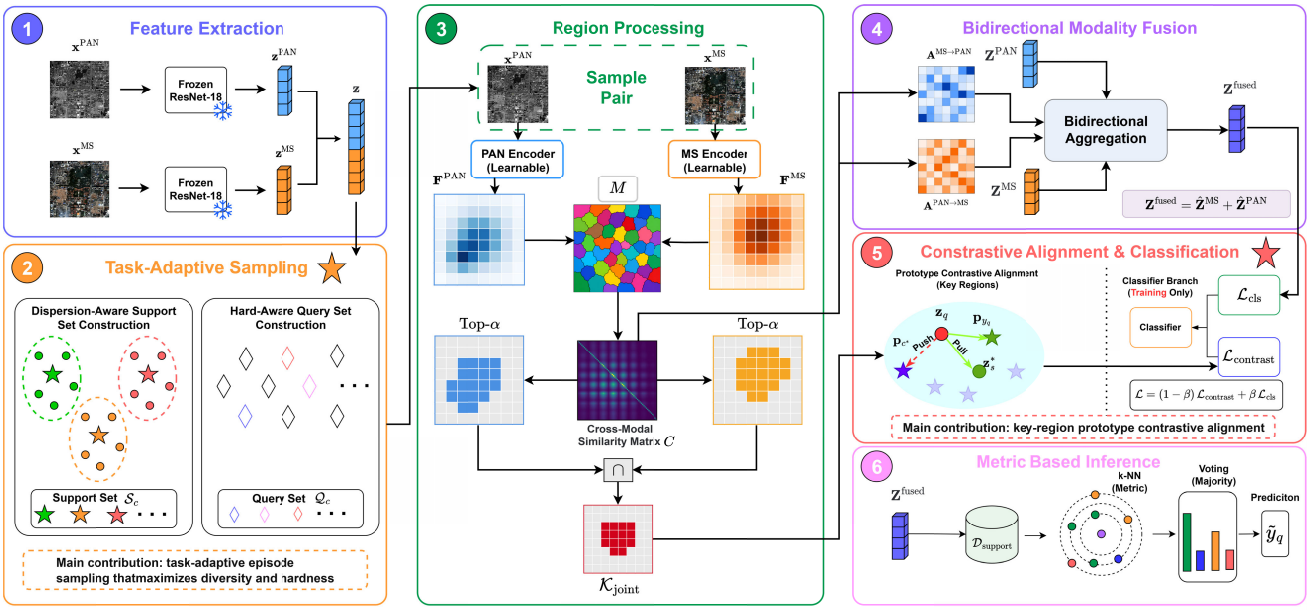



Fig. 1. Overall framework of the proposed method. The pipeline consists of three main phases: 1) extracting global prior features to perform TAS for forming informative support and query sets; 2) extracting region-level spatial features and mining a cross-modal consensus mask via the KMA module; and 3) utilizing the key-region mask for contrastive alignment and bidirectional cross-modal spatial attention for complementary feature fusion. During inference, a nonparametric $k$-NN classifier is used to evaluate the query based on the few-shot support reference bank.

[34] jointly model overarching contexts and fine-grained local features to enhance cross-modal semantic consistency. Other studies focus on structural integration; for example, dynamic multiscale fusion frameworks [35] strengthen multisource representation learning via hierarchical feature aggregation. Furthermore, shared-private decoupling architectures [36] perform multilevel feature alignment, effectively isolating modality-specific noise while extracting invariant cross-modal semantics. Related remote sensing studies have explored Mamba-based spectral representation learning and multimodal semantic understanding, respectively [37], [38]. Recent remote sensing visual understanding studies have also explored instance-level region perception and referring-region grounding with foundation models [39], [40]. Although these studies address different tasks, they reflect the increasing importance of region-aware semantic modeling in remote sensing.

Although these methods demonstrate the effectiveness of global or semantic-level alignment, they mainly operate at coarse feature scales and do not explicitly model locally discriminative regions critical for classification. In data-scarce few-shot scenarios, the impacts of background interference and regional misalignment are amplified, significantly limiting the robustness of cross-modal feature fusion. Motivated by this limitation, this article focuses on key-region-level cross-modal alignment to explicitly suppress background noise and enhance discriminative learning with limited labeled samples.

## III. Proposed Method

### A. Overall Framework

TK-Net follows a task-construction–region-alignment –metric-inference workflow, as illustrated in Fig. 1. TAS constructs informative episodes, KMA mines key regions for selective contrastive alignment and performs cross-modal feature fusion, and a nonparametric $k$-NN classifier is used during inference.

The complete workflow consists of the following stages.

1) *Feature Extraction:* Global feature representations ($\mathbf{z}^{\mathrm{PAN}}, \mathbf{z}^{\mathrm{MS}}$) are extracted for the entire dataset using initialized ResNet-18 encoders and serve as a static prior for TAS.
2) *TAS:* Dispersion-aware support set selection and hard-aware query set construction form informative episodes that enhance prototype robustness and sharpen decision boundaries.
3) *Region Processing:* A shared superpixel map $\mathcal{M}$ and cross-modal consistency scores are computed to extract the key region set ($\mathcal{K}_{\mathrm{joint}}$), which acts as a spatial mask to eliminate background interference.
4) *Modality Fusion:* Bidirectional spatial attention aggregates complementary PAN and MS region features, yielding a fused representation ($\mathbf{Z}^{\mathrm{fused}}$).
5) *Contrastive Alignment and Classification:* A region-selective contrastive loss ($\mathcal{L}_{\mathrm{contrast}}$) aligns prototypes within key regions, and a cross-entropy loss ($\mathcal{L}_{\mathrm{cls}}$) trains the classifier on fused features. The overall loss is

$$\mathcal{L} = (1-\beta)\,\mathcal{L}_{\mathrm{contrast}} + \beta\,\mathcal{L}_{\mathrm{cls}} \tag{1}$$

where $\beta \in [0, 1]$ balances the two objectives.
6) *Metric-Based Inference:* During evaluation, a nonparametric $k$-NN classifier predicts the query class by retrieving the most similar examples from the few-shot reference bank.

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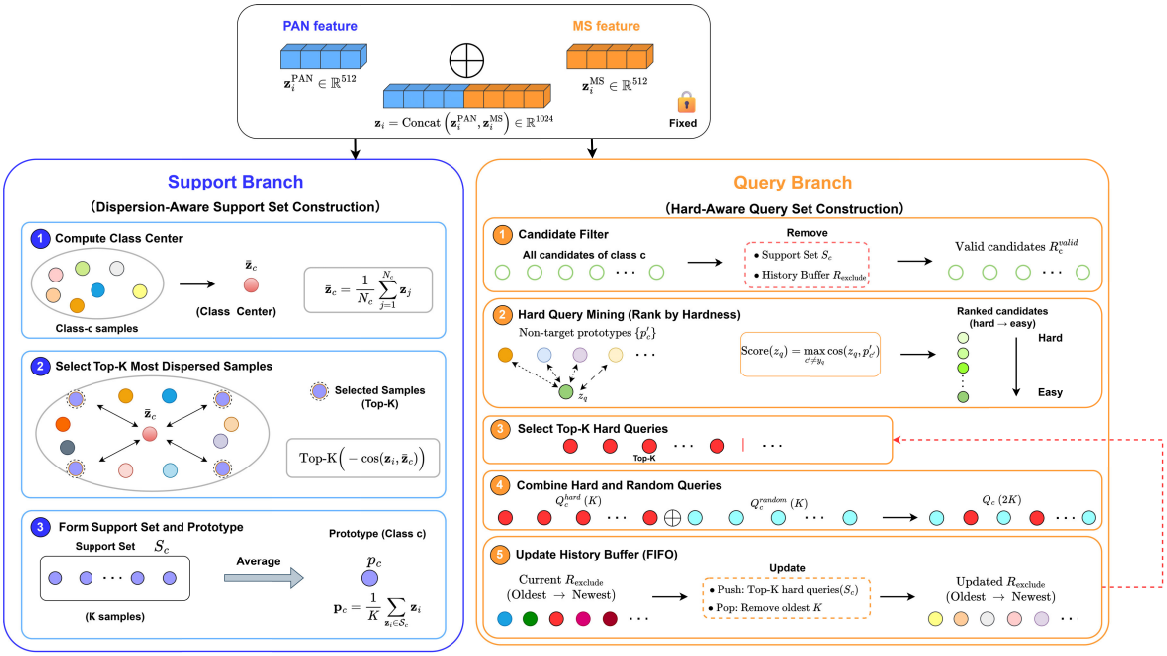



Fig. 2. Overview of the proposed TAS module. For each class, we compute the class center and select the most dispersed samples (with the lowest similarity to the center) to form the support set. Based on the resulting prototypes, we then identify hard query candidates by measuring their similarity to prototypes of other classes. Hard and randomly sampled queries are combined, and a buffer filter is applied to prevent overuse of recent samples. The final query set is constructed from the remaining candidates.

### B. TAS Module

In traditional FSL, support and query sets are typically sampled randomly, which may yield overly simple tasks. To address this limitation, we propose the TAS module, which employs a feature-driven sampling strategy to construct more informative and discriminative tasks. The TAS module consists of three sequential steps: feature extraction, dispersion-aware support set selection, and hard-aware query set construction. The complete workflow is illustrated in Fig. 2.

*1) Feature Extraction:* To facilitate feature-driven task construction, we begin by establishing a global feature space for the entire training set. To ensure computational efficiency, this extraction is performed as a one-off process before episodic training, utilizing the initialized encoders. This avoids the significant overhead of repetitive re-extraction during training iterations.

Specifically, we employ two modality-specific ResNet-18 encoders, $f_{\text{PAN}}$ and $f_{\text{MS}}$, to process the PAN ($x^{\text{PAN}}$) and MS ($x^{\text{MS}}$) inputs, respectively. Here, the PAN input follows the same input-level alignment as described in the KMA module, i.e., it is downsampled to the MS spatial grid before being fed into the PAN encoder. To obtain compact global signatures $\mathbf{z} \in \mathbb{R}^{512}$ suitable for efficient sampling, we apply adaptive average pooling directly to the encoder feature maps

$$\begin{aligned} \mathbf{z}^{\text{PAN}} &= \text{AdaptiveAvgPool}\left(f_{\text{PAN}}\left(x^{\text{PAN}}\right)\right) \\ \mathbf{z}^{\text{MS}} &= \text{AdaptiveAvgPool}\left(f_{\text{MS}}\left(x^{\text{MS}}\right)\right). \end{aligned} \tag{2}$$

These precomputed 512-D vectors remain fixed throughout the training process. They serve solely as the geometric basis for measuring sample dispersion and difficulty in the subsequent TAS module, independent of the learnable region-level representations used for classification.

*2) Dispersion-Aware Support Set Construction:* To construct diversity-aware class prototypes that cover relatively dispersed in-class variations, we propose a dispersion-aware strategy for support set construction.

First, to ensure both modalities jointly inform the sampling, we generate a unified representation $\mathbf{z}_i = \text{Concat}(\mathbf{z}_i^{\text{PAN}}, \mathbf{z}_i^{\text{MS}}) \in \mathbb{R}^{1024}$ by concatenating the modality-specific features along the channel dimension.

Next, we identify the support samples that are most distinct from the global class center. For a specific class $c$, the support set $\mathcal{S}_c$ is formed by selecting the $K$ samples with the lowest similarity to the class mean

$$\mathcal{S}_c = \underset{i}{\text{Top-}K}\left(-\frac{\mathbf{z}_i \cdot \bar{\mathbf{z}}_c}{||\mathbf{z}_i||||\bar{\mathbf{z}}_c||}\right) \tag{3}$$

where $\bar{\mathbf{z}}_c = 1/N_c \sum_{j=1}^{N_c} \mathbf{z}_j$ denotes the global mean vector calculated over all $N_c$ available samples for class $c$. The negative sign within the Top-$K$ operator gives higher priority to labeled samples with lower similarity to the class center, encouraging the support set to cover relatively diverse in-class variations rather than only central samples.

Finally, we compute the class prototype $\mathbf{p}_c$ by averaging these dispersed samples

$$\mathbf{p}_c = \frac{1}{K} \sum_{\mathbf{z}_i \in \mathcal{S}_c^{\text{support}}} \mathbf{z}_i. \tag{4}$$

The resulting prototype $\mathbf{p}_c$ subsequently serves as the reference for query classification. Unlike random sampling, this strategy prioritizes relatively dispersed labeled samples to reduce the bias toward only central instances. It is intended to

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improve the diversity of the support set, rather than to identify or deliberately select outliers.

*3) Hard-Aware Query Set Construction:* To expose the model to challenging decision boundaries and encourage the learning of discriminative features, we construct a hard-aware query set by explicitly mining confusing instances.

The detailed workflow of this construction operates in a "filter-then-rank" manner, as visually summarized in Fig. 2. First, to prevent overfitting to specific hard examples and ensure temporal diversity, we maintain a history buffer $\mathcal{R}_{\text{exclude}}$ that records samples selected in recent episodes. Valid query candidates, denoted as $\mathcal{R}_c^{\text{valid}}$, are identified by excluding both the current support set and any samples currently stored in this buffer. Next, we quantify the "hardness" of each remaining candidate based on its proximity to interfering classes. For a candidate sample $\mathbf{z}_q$ with ground-truth label $y_q$, we define its hardness score as the maximum cosine similarity to the prototype of any incorrect class

$$\text{Score}\left(\mathbf{z}_q\right) = \max_{c' \neq y_q} \cos\left(\mathbf{z}_q, \mathbf{p}_{c'}\right) \tag{5}$$

where $\mathbf{p}_{c'}$ denotes the prototype of a nontarget class $c'$. A higher score indicates that the sample lies closer to the decision boundary of an interfering category, making it a high-value training instance.

According to the obtained hardness scores, the $K$ highest-ranked samples are selected as the hard query subset $\mathcal{Q}_c^{\text{hard}}$. This subset is then combined with randomly sampled instances $\mathcal{Q}_c^{\text{random}}$ to constitute the final query set $\mathcal{Q}_c$, maintaining a balanced distribution. Finally, the selected hard samples are pushed into $\mathcal{R}_{\text{exclude}}$ using a first-in-first-out strategy. This rolling update mechanism ensures that the exclusion constraint adapts dynamically as training progresses.

Unlike random sampling strategies that treat all instances equally, this module prioritizes hard samples located near the decision boundaries. By forcing the model to distinguish the target class from its most confusing competitors, it generates denser and more informative supervisory signals. This explicit hard mining prevents the network from converging on trivial, suboptimal decision boundaries, thereby enhancing its cross-modal discriminative capability.

Overall, the TAS module enables a feature-driven construction of few-shot tasks, where both support and query sampling strategies are explicitly optimized. By incorporating feature dispersion and instance difficulty, our approach lays a solid foundation for subsequent classification and alignment stages.

### C. KMA Module

Due to the inherent spectral–spatial discrepancies between PAN and MS modalities, directly aligning or fusing entire image patches is highly suboptimal. Global aggregation indiscriminately blends target objects with task-irrelevant background noise, diluting the discriminative foreground features and yielding inaccurate class prototypes. To address this, we propose the KMA

module. Rather than performing unconstrained global alignment, KMA leverages superpixel-guided cross-modal consensus to isolate informative foreground regions explicitly. By restricting prototype-level contrastive learning exclusively to these key areas, the module effectively mitigates background interference. As illustrated in Fig. 3, KMA comprises four components: region processing, contrastive prototype alignment, modality fusion, and metric-based inference.

*1) Region Processing:* To obtain spatially consistent region-level features for cross-modal alignment, we adopt a shared segmentation strategy. Each MS patch ($16 \times 16$) and its corresponding PAN patch ($64 \times 64$) cover the same geographic extent. The PAN patch is bilinearly downsampled to $16 \times 16$ before feature extraction. Both aligned patches are fed into modality-specific encoders with stride 1, producing feature maps with matched spatial dimensions. A shared superpixel map $\mathcal{M}$ is then generated by concatenating the aligned PAN and MS inputs along the channel dimension and applying SLIC segmentation. For a given image pair, the modality-specific encoders output dense spatial feature maps $\mathbf{F}^m$. Based on $\mathcal{M}$, the region-level feature representation for the $k$th superpixel is extracted via average pooling

$$\mathbf{z}_k^m = \text{AvgPool}\left(\mathbf{F}^m, \mathcal{M}_k\right), \quad m \in \{\text{PAN}, \text{MS}\} \tag{6}$$

where $\mathcal{M}_k$ indicates the spatial coordinates of the $k$th superpixel. This shared mask ensures that region tokens from both modalities have identical spatial partitions, enabling direct cross-modal correspondence modeling.

To eliminate background noise and focus on discriminative foreground objects, we selectively identify a subset of key regions based on cross-modal consensus. Intuitively, key regions correspond to structures that are consistently informative across both modalities, such as building edges, road segments, or vegetation textures, which provide more reliable anchors for alignment than homogeneous background areas. We compute a cross-modal similarity matrix $\mathbf{C} \in \mathbb{R}^{N \times N}$ to quantify region-wise correspondences

$$\mathbf{C}_{ij} = \cos\left(\mathbf{z}_i^{\text{MS}}, \mathbf{z}_j^{\text{PAN}}\right). \tag{7}$$

The importance scores of MS and PAN regions are then computed by averaging row and column similarities

$$S_i^{\text{MS}} = \frac{1}{N}\sum_{j=1}^{N} \mathbf{C}_{ij}, \quad S_j^{\text{PAN}} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{C}_{ij} \tag{8}$$

where $N$ is the total number of superpixels. We select the top $\alpha$ proportion of regions from each modality to form candidate sets $\mathcal{K}_{\text{MS}}$ and $\mathcal{K}_{\text{PAN}}$. The final key region set $\mathcal{K}_{\text{joint}}$ is defined as their intersection

$$\mathcal{K}_{\text{joint}} = \text{Top-}\alpha\left(S^{\text{MS}}\right) \cap \text{Top-}\alpha\left(S^{\text{PAN}}\right). \tag{9}$$

Rather than utilizing all superpixels for contrastive learning, which may introduce irrelevant background information, we restrict the contrastive constraint to $\mathcal{K}_{\text{joint}}$. Specifically, $\mathcal{K}_{\text{joint}}$ serves as a spatial mask for region-selective prototype contrastive alignment, while feature fusion is performed through bidirectional cross-modal spatial attention.

*2) Contrastive Prototype Alignment:* In this module, we perform cross-modal alignment within the previously selected key regions using a prototype-based contrastive learning strategy. To strengthen the supervision signal, we treat each query

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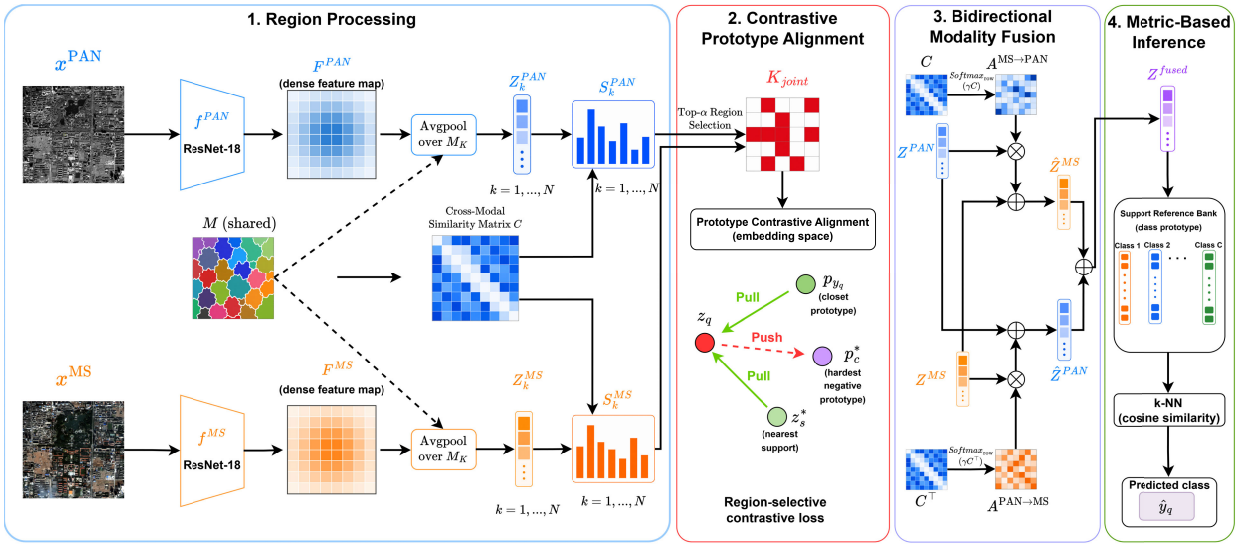



Fig. 3. Detailed architecture of the KMA module. Modality-specific encoders extract dense feature maps, which are converted into region-level tokens through shared SLIC superpixel segmentation and average pooling. A cross-modal similarity matrix is computed to mine the joint key-region set $\mathcal{K}_{\text{joint}}$ and derive bidirectional spatial attention for feature fusion. The key regions serve as a spatial constraint for prototype contrastive alignment, and the fused representation is evaluated by $k$-NN over the support reference bank.

region from one modality as an anchor and construct cross-modal positive and negative samples from the other modality.

*a) Hybrid positive matching:* For a selected key query region $\mathbf{z}_q$ with class label $y_q$, relying solely on the class prototype may ignore specific instance details, while relying only on nearest neighbors can be unstable. Therefore, we define a hybrid positive similarity $S_{\text{pos}}$ that integrates both global and local semantics

$$S_{\text{pos}}\left(\mathbf{z}_q\right)=\frac{1}{2}\left(\cos\left(\mathbf{z}_q,\mathbf{p}_{y_q}\right)+\max_{\mathbf{z}_s\in\mathcal{S}_{y_q}}\cos\left(\mathbf{z}_q,\mathbf{z}_s\right)\right) \tag{10}$$

where $\mathbf{p}_{y_q}$ denotes the prototype of the ground-truth class, and $\mathcal{S}_{y_q}$ represents the set of support samples belonging to class $y_q$. The first term pulls the query toward the class center, while the second term identifies the nearest support sample, denoted as $\mathbf{z}_s^*=\arg\max_{\mathbf{z}_s\in\mathcal{S}_{y_q}}\cos(\mathbf{z}_q,\mathbf{z}_s)$. This formulation effectively balances intraclass generalization with fine-grained structural preservation.

*b) Hard negative mining:* To enforce sharp decision boundaries, we employ a hard negative mining strategy. The negative similarity $S_{\text{neg}}$ is defined as the maximum similarity to any incorrect class prototype

$$S_{\text{neg}}\left(\mathbf{z}_q\right)=\max_{c\neq y_q}\left(\cos\left(\mathbf{z}_q,\mathbf{p}_c\right)\right). \tag{11}$$

Instead of pushing away all incorrect classes equally, this formulation forces the model to concentrate its optimization efforts on distinguishing the target class from its most confusing semantic competitor. We denote this hardest negative prototype as $\mathbf{p}_{c^*}$, where $c^*=\arg\max_{c\neq y_q}\cos(\mathbf{z}_q,\mathbf{p}_c)$.

*c) Region-selective loss formulation:* To eliminate the interference of noise, the contrastive loss is computed exclusively on the intersection set $\mathcal{K}_{\text{joint}}$ obtained from the key region mining module (hereafter denoted as $\mathcal{K}$ for notational brevity). The region-wise margin ranking loss for a single direction (e.g., MS to PAN) is formulated as

$$\mathcal{L}_{\text{MS}\to\text{PAN}}=\frac{1}{|\mathcal{K}|}\sum_{q\in\mathcal{K}}\max\left(0,\Delta+S_{\text{neg}}\left(\mathbf{z}_q^{\text{MS}}\right)-S_{\text{pos}}\left(\mathbf{z}_q^{\text{MS}}\right)\right) \tag{12}$$

where $\Delta$ is a fixed margin. Symmetrically, the reverse alignment loss $\mathcal{L}_{\text{PAN}\to\text{MS}}$ is computed by swapping the roles of the two modalities and using $\mathbf{z}_q^{\text{PAN}}$ as anchors. The total contrastive loss is defined as the sum of these bidirectional losses

$$\mathcal{L}_{\text{contrast}}=\mathcal{L}_{\text{MS}\to\text{PAN}}+\mathcal{L}_{\text{PAN}\to\text{MS}}. \tag{13}$$

By restricting the optimization to $\mathcal{K}$ and employing the hybrid sampling strategy, this formulation provides a robust supervisory signal driven solely by high-confidence, cross-modally consistent foreground regions.

*3) Bidirectional Modality Fusion:* In this module, we perform complementary PAN–MS feature fusion using bidirectional cross-modal spatial attention. Let $\mathbf{Z}^{\text{MS}}\in\mathbb{R}^{N\times d}$ and $\mathbf{Z}^{\text{PAN}}\in\mathbb{R}^{N\times d}$ denote the region-level feature matrices of the MS and PAN modalities, respectively. Based on the same cross-modal similarity matrix $\mathbf{C}$ defined above, row-wise Softmax is applied to $\mathbf{C}$ and its transpose to obtain two directional attention matrices

$$\begin{aligned}\mathbf{A}^{\text{MS}\to\text{PAN}}&=\text{Softmax}_{\text{row}}\left(\gamma\mathbf{C}\right)\\ \mathbf{A}^{\text{PAN}\to\text{MS}}&=\text{Softmax}_{\text{row}}\left(\gamma\mathbf{C}^{\top}\right)\end{aligned} \tag{14}$$

where $\gamma$ is a scaling factor. $\mathbf{A}^{\text{MS}\to\text{PAN}}$ aggregates PAN region features for each MS region, while $\mathbf{A}^{\text{PAN}\to\text{MS}}$ aggregates MS region features for each PAN region. The bidirectionally enhanced features are computed as

$$\begin{aligned}\hat{\mathbf{Z}}^{\text{MS}}&=\mathbf{Z}^{\text{MS}}+\mathbf{A}^{\text{MS}\to\text{PAN}}\mathbf{Z}^{\text{PAN}}\\ \hat{\mathbf{Z}}^{\text{PAN}}&=\mathbf{Z}^{\text{PAN}}+\mathbf{A}^{\text{PAN}\to\text{MS}}\mathbf{Z}^{\text{MS}}\\ \mathbf{Z}^{\text{fused}}&=\hat{\mathbf{Z}}^{\text{MS}}+\hat{\mathbf{Z}}^{\text{PAN}}.\end{aligned} \tag{15}$$

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This formulation does not perform explicit modality-wise selection for each region. Instead, it models cross-modal region correspondence and aggregates complementary information from semantically related regions in the counterpart modality, while preserving the original modality-specific representation through residual connections.

*4) Metric-Based Inference:* For evaluation, we employ a nonparametric $k$-nearest neighbors classifier. We construct a few-shot reference bank $\mathcal{D}_{\text{support}}$ using the fused features of the limited labeled support samples. During inference, for each test query $\mathbf{z}_q$, we retrieve its $k$ nearest neighbors from $\mathcal{D}_{\text{support}}$ based on cosine similarity

$$\mathcal{N}_k\left(\mathbf{z}_q\right) = \underset{(\mathbf{z}_i, y_i)\in\mathcal{D}_{\text{support}}}{\text{Top-}k} \cos\left(\mathbf{z}_q, \mathbf{z}_i\right). \tag{16}$$

The final class prediction is determined via majority voting within $\mathcal{N}_k(\mathbf{z}_q)$. In our experiments, we set $k = 5$. In summary, this cross-modal framework reduces modality discrepancies through region-selective alignment and adaptive fusion. Unlike traditional methods that align all spatial locations indiscriminately, our approach strictly limits prototype-based contrastive optimization to key regions with strong semantic consensus. Furthermore, the hybrid positive matching and hard negative mining effectively capture both typical intraclass variations and challenging boundary-level semantics. By doing so, the model achieves enhanced structural consistency and learns highly discriminative, modality-invariant representations.

## IV. Experimental Results

### A. Dataset Description and Experimental Environment

We evaluate TK-Net on three dual-source remote sensing benchmarks, including the Hohhot, Nanjing, and Xi'an datasets. The main properties of each dataset are summarized below.

*1) Hohhot Dataset:* This dataset provides paired MS and PAN images. The MS image has a spatial resolution of 3.2 m and a size of $2001 \times 2101 \times 4$, while the corresponding PAN image has a finer spatial resolution of 0.8 m with a size of $8004 \times 8404 \times 1$. It contains 11 land-cover categories, including water, road, bare land, vegetation, farmland, and six types of building areas.

*2) Nanjing Dataset:* The Nanjing dataset also consists of paired MS and PAN observations. The MS image has a spatial resolution of 3.2 m and a size of $2000 \times 2500 \times 4$, while the PAN image has a spatial resolution of 0.8 m with a size of $8000 \times 10\,000 \times 1$. The annotations cover 11 land-cover categories, including water, bare land, two types of roads, two types of vegetation, and five types of building areas.

*3) Xi'an Dataset:* The Xi'an dataset contains one MS image and its corresponding PAN image. The MS image has a spatial resolution of 3.2 m with a size of $4541 \times 4548 \times 4$, whereas the PAN image has a spatial resolution of 0.8 m and a size of $18\,164 \times 18\,192 \times 1$. This dataset is labeled into 12 land-cover classes, including water, bare land, farmland, two road categories, two vegetation categories, and five building-area categories.

All experiments are implemented using the PyTorch deep learning framework and executed on a workstation equipped with a single NVIDIA 3090 GPU.

### B. Experimental Setup

*1) Evaluation Metrics and Input Configuration:* To assess the classification results, we report three standard metrics, including overall accuracy (OA), average accuracy (AA), and the Kappa coefficient ($\kappa$). Considering that the spatial resolution of PAN images is four times higher than that of MS images, the two modalities are cropped into paired patches with matched geographic coverage. Specifically, each MS patch is set to $16 \times 16 \times 4$, and the corresponding PAN patch is set to $64 \times 64 \times 1$, thereby preserving spatial consistency between the paired inputs.

*2) Few-Shot Task Protocol:* Unlike traditional fully supervised methods that rely on abundant annotated data, our experiments are strictly conducted under few-shot settings. During the training phase, we adopt an episodic paradigm, where each task is formulated as a 5-way 1-shot classification problem. Specifically, in each episode, we randomly sample five distinct categories from the total dataset classes, and for each selected category, we only use one labeled support sample. The query set contains ten samples per class.

This setting is adopted for two main reasons. First, the 5-way 1-shot protocol represents a challenging and widely used few-shot evaluation setting, which better reflects the practical scenario in which only very limited labeled samples are available for each land-cover category. Second, using ten query samples per class provides sufficient query instances to form stable episodic supervision while maintaining task difficulty. To further expose the model to challenging decision boundaries, the query set is constructed by dynamically mixing hard and random candidates, with a 30% ratio of hard query samples.

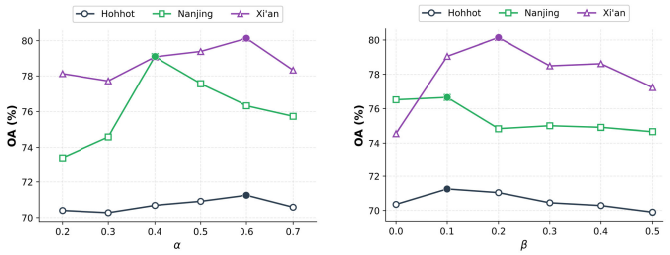
During the evaluation phase, to prevent data leakage and ensure a rigorous assessment, we randomly sample a highly restricted portion of the dataset (i.e., 2% data ratio) to construct the few-shot support reference bank $\mathcal{D}_{\text{support}}$. The nonparametric $k$-NN classifier ($k = 5$) is then applied to evaluate all the remaining extensive test samples.

### C. Comparison and Analysis of Hyperparameters

In this section, we mainly make a detailed comparison and analysis of two important hyperparameters in our network: the top-$k$ ratio $\alpha$ that controls the proportion of superpixels selected by the cross-modal attention mask, and the tradeoff coefficient $\beta$ between the cross-entropy and contrastive losses. Because the two hyperparameters are independent, we use the control variable method to analyze their selection. Although the actual tuning was conducted in the order $\beta$ then $\alpha$, we present the analysis in the order in which the two hyperparameters are introduced, which does not affect the control-variable comparison.

We vary $\alpha$ from 0.2 to 0.7 with a stride of 0.1, while fixing $\beta$ at 0.1 (Hohhot), 0.1 (Nanjing), and 0.2 (Xi'an). The corresponding results are shown in Fig. 4(b). It can be seen

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(a) (b)

Fig. 4. (a) Results of different values of hyperparameters $\alpha$. (b) Results of different values of hyperparameters $\beta$.

that performance is best when $\alpha$ equals 0.6 (Hohhot and Xi'an) and 0.4 (Nanjing), and when $\alpha$ is away from these values, OA decreases. When $\alpha$ is too small, only a few highly confident superpixels are selected as key regions, potentially discarding informative regions and weakening the contrastive supervision signal. When $\alpha$ is too large, too many superpixels are included, introducing background noise that dilutes the cross-modal alignment and reduces feature discriminability. The optimal $\alpha$ therefore reflects a dataset-dependent tradeoff between region selectivity and spatial coverage in the KMA module.

We vary $\beta$ from 0 to 0.5 with a stride of 0.1, while fixing the top-$k$ ratio at 0.6. The corresponding results are shown in Fig. 4(b). It can be seen that, in each dataset, performance is best when $\beta$ equals 0.1 (Hohhot), 0.1 (Nanjing), and 0.2 (Xi'an), and that when $\beta$ is away from these values, OA decreases. When $\beta$ is too large, the CE loss dominates training, and contrastive supervision at the superpixel level is weakened, degrading the feature quality required for KNN-based evaluation. When $\beta$ is too small (or zero), the model relies solely on contrastive learning, which may benefit smaller datasets but lacks global classification guidance for larger, more complex scenes. The optimal $\beta$ therefore reflects a dataset-dependent balance between the two objectives.

### D. Module Comparison Experiments

To further verify the effectiveness of the proposed modules, we conduct detailed qualitative analyses on the Hohhot dataset, maintaining the identical backbone and hyperparameter settings for fairness.

*1) Validation of TASs:* To further analyze the effectiveness of the proposed TAS strategy, we conduct an episode-level sampling mechanism analysis under the same setting as model training, i.e., 5-way 1-shot 10-query. Instead of relying on subjective visual differences among image patches, we evaluate how TAS changes the informativeness of constructed episodes in the learned feature space.

As shown in Fig. 5(a), TAS selects support samples with significantly larger distances from their corresponding class centroids than random sampling. This indicates that the dissimilar support selection strategy tends to choose more peripheral samples, which helps the 1-shot support set better capture intraclass variations. Fig. 5(b) further compares query hardness, measured by the cosine distance from each query sample to the support prototype. Here, Random Q denotes queries from random sampling, TAS Hard Q denotes the hard queries explicitly selected by TAS, and TAS Fill Q denotes the remaining queries randomly filled to keep the same 10-query setting. TAS Hard Q shows much larger distances than Random Q, while TAS Fill Q maintains moderate hardness and provides additional query diversity. This indicates that TAS constructs more challenging yet balanced query sets for episodic training.

In addition, Fig. 5(c) evaluates the sample reuse behavior of hard queries. By maintaining a recently sampled set, TAS reduces repeated selection of the same hard instances and encourages a more balanced exploration of difficult samples. Quantitatively, TAS increases support extremeness by 4.23 times and hard-query difficulty by 3.63 times compared with random sampling, while reducing the maximum hard-query reuse rate by 19.2%. These results verify that TAS improves episodic task construction from three aspects: more informative support selection, harder query sampling, and reduced hard-sample redundancy.

*2) Validation of KMA:* To validate the KMA module's feature enhancement capabilities, we visualize the spatial activations of the fused features. For a fair and unified evaluation, we adopt a structure-aware decomposition, namely PCA-guided principal component analysis weighted by Harris corner responses, to map high-dimensional tensors into 2-D heatmaps. As shown in Fig. 6(b), the baseline method (element-wise summation) activates diffusely across the entire patch, failing to distinguish semantically meaningful structures from task-irrelevant background noise. Under KMA modulation [Fig. 6(c)], feature activations become sharply concentrated on discriminative structural boundaries, such as building edges and road networks. This spatial refinement is further corroborated by the difference maps [Fig. 6(d)], where the pronounced red regions (indicating KMA-induced improvements) are strictly localized along the target object contours rather than uniformly distributed. This pixel-level evidence validates that KMA effectively utilizes cross-modal consensus to suppress background interference and amplify structurally relevant semantics.

*3) Ablation Study:* In this section, we conduct a series of ablation experiments to analyze the contribution and effectiveness of each component in the proposed framework. All experiments are conducted on the Hohhot dataset, with ResNet-18 serving as the baseline. The quantitative results, including OA, AA, and Kappa coefficient, are summarized in Table I.

Table I shows that each proposed component consistently yields performance gains. Integrating attention fusion and TAS improves the OA from 67.62% to 69.06%, confirming their ability to refine cross-modal features and suppress noise. The introduction of contrastive alignment leads to more significant improvements. Specifically, global CL increases the OA to 70.48%, while our proposed key region CL achieves the best performance at 71.25% (a 3.63% total improvement over the baseline). These results demonstrate that aligning fine-grained discriminative regions is more effective than global alignment in eliminating modal differences and enhancing overall fusion

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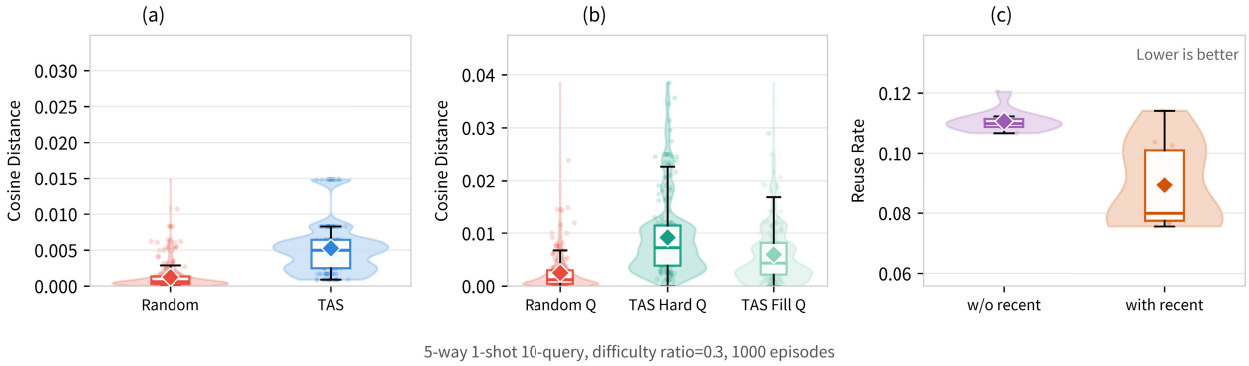



Fig. 5. Episode-level sampling mechanism analysis of the proposed TAS strategy under the 5-way 1-shot 10-query setting. The three panels evaluate (a) support extremeness, (b) query hardness, and (c) hard-query reuse, respectively.

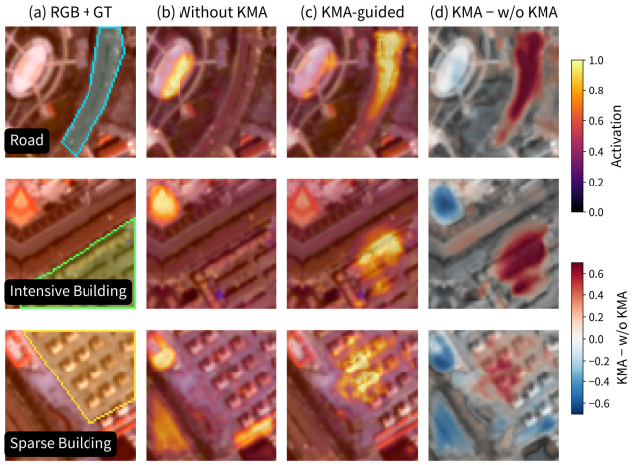



Fig. 6. Visual ablation of the KMA module using structure-aware feature decomposition. (a) RGB patches. (b) Baseline features. (c) KMA-guided features. (d) Difference maps.

TABLE I
ABLATION STUDY OF DIFFERENT COMPONENTS ON THE PROPOSED FRAMEWORK

| Baseline | Attn Fusion | TAS | Contrastive Alignment | | OA (%) | AA (%) | Kappa (%) |
|---|---|---|---|---|---|---|---|
| | | | Global CL | Key Reg. CL | | | |
| ✓ | | | | | 67.62 | 68.09 | 63.22 |
| ✓ | ✓ | | | | 68.41 | 69.54 | 64.18 |
| ✓ | ✓ | ✓ | | | 69.06 | 70.28 | 64.91 |
| ✓ | ✓ | ✓ | ✓ | | 70.48 | 70.73 | 66.41 |
| ✓ | ✓ | ✓ | | ✓ | **71.25** | **71.23** | **67.31** |

quality. This comparison directly verifies the contribution of KMA, since the performance gain from global CL to key region CL comes from restricting the alignment scope to the mined key regions.

Figs. 7–9 arrange the six methods in columns. For each method, the rows show predictions at labeled test locations and a full-scene map paired with an enlarged view; dashed lines link each outlined region to its enlargement. This layout preserves scene-level context while making local classification patterns easier to compare.

TABLE II
QUANTITATIVE EVALUATION RESULTS ON THE HOHHOT DATASET

| Methods | Cross CLMP [24] | SLCL [23] | NNC-Net [41] | I$^3$RL-Net [42] | CMS$^2$I-Mamba [43] | TK-Net (Ours) |
|---|---|---|---|---|---|---|
| $c_1$ | **94.22** | 93.07 | 93.70 | 80.47 | 93.30 | 87.17 |
| $c_2$ | 56.39 | 67.73 | 74.41 | 55.60 | **81.41** | 73.57 |
| $c_3$ | 90.36 | 96.10 | 41.75 | **96.85** | 81.06 | 84.16 |
| $c_4$ | **83.59** | 52.97 | 79.41 | 50.10 | 72.55 | 81.18 |
| $c_5$ | 52.35 | 40.75 | 62.46 | 46.04 | 48.21 | **74.76** |
| $c_6$ | 38.64 | 64.34 | **70.74** | 68.47 | 27.99 | 62.86 |
| $c_7$ | 51.64 | 53.49 | 71.55 | 20.39 | 32.37 | **77.14** |
| $c_8$ | 82.93 | 83.81 | **95.46** | 62.23 | 87.62 | 85.83 |
| $c_9$ | 54.97 | 48.82 | 55.26 | 51.85 | **60.05** | 48.16 |
| $c_{10}$ | 23.09 | 8.69 | 46.48 | 45.96 | 21.65 | **51.62** |
| $c_{11}$ | 45.10 | 27.14 | 38.05 | 50.74 | 45.07 | **57.15** |
| OA (%) | 58.74 | 59.96 | 66.62 | 56.53 | 57.22 | **71.25** |
| AA (%) | 61.21 | 57.90 | 66.30 | 57.15 | 59.21 | **71.23** |
| Kappa (%) | 53.32 | 54.66 | 62.12 | 50.81 | 51.42 | **67.31** |

### E. Experimental Result on Hohhot Dataset

In this section, we take the Hohhot dataset as an example to compare various methods and verify in detail the effectiveness of the proposed TK-Net. The specific quantitative results are shown in Table II.

Among the comparison methods, CrossCLMP employs a contrastive pretraining strategy with inter- and intrasource alignment to learn transferable representations for few-shot classification. However, its reliance on large-scale unlabeled pretraining data limits its effectiveness when the domain gap between pretraining and downstream data is substantial, resulting in OA of 58.74%. SLCL improves upon CrossCLMP with a sample weighting and screening strategy to mitigate false negatives in contrastive learning, along with a hard sample learning module. This yields modest improvements (OA 59.96%), but the lack of explicit cross-modal feature fusion still constrains its discriminative power on complex building categories.

NNCNet, originally designed for HSI-LiDAR classification, adopts a nearest neighbor-based data augmentation within a MoCo contrastive framework and employs bilinear attention for second-order cross-modal interaction. This strong contrastive pretraining enables it to achieve the highest OA (66.62%) among all comparison methods, with particularly competitive performance on vegetation (c4, 79.41%) and water (c1, 93.70%). However, its bilinear attention is

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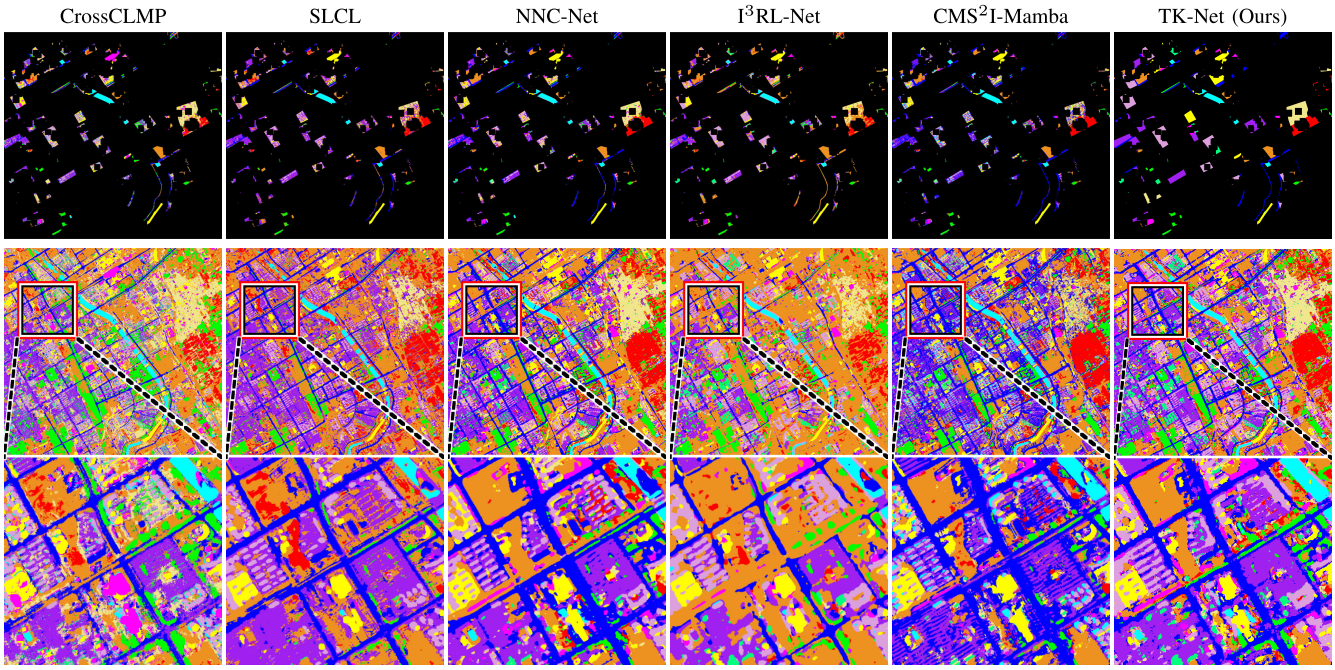



Fig. 7. Classification maps and local comparisons on the Hohhot dataset. The columns show CrossCLMP, SLCL, NNC-Net, I$^3$RL-Net, CMS$^2$I-Mamba, and TK-Net, respectively. (Top to bottom) Rows show predictions at labeled test locations and full-scene maps paired with enlarged local views; dashed lines connect the selected regions to their enlargements.

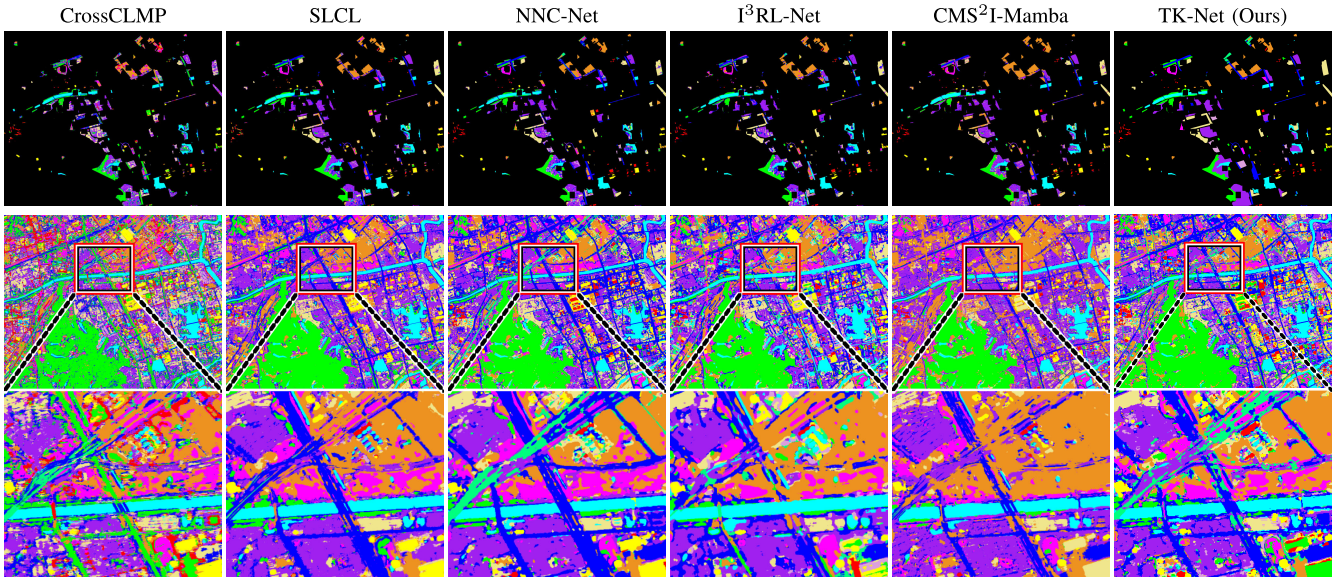



Fig. 8. Classification maps and local comparisons on the Nanjing dataset. The columns show CrossCLMP, SLCL, NNC-Net, I$^3$RL-Net, CMS$^2$I-Mamba, and TK-Net, respectively. (Top to bottom) Rows show Predictions at labeled test locations and full-scene maps paired with enlarged local views; dashed lines connect the selected regions to their enlargements.

tailored to HSI-LiDAR modalities, which limits its adaptability to the PAN-MS resolution gap, resulting in poor performance on certain building categories such as red building (c10, 46.48%).

I$^3$RL-Net proposes a cross-source interactive enhanced fusion module and a complementary representation supervised learning strategy to reduce intersource redundancy. Despite these designs, its OA of 56.53% indicates that standard supervised fusion without contrastive alignment struggles under the few-shot setting of Hohhot.

CMS$^2$I-Mamba leverages the Mamba state-space model for efficient long-range dependency modeling and introduces a channel interaction alignment module for cross-modal semantic alignment. While it excels on road (c2, 81.41%) due to its global spatial modeling capacity, its overall OA of 57.22% suggests that Mamba-based modeling alone is insufficient to capture the fine-grained spectral–spatial correspondence required for subcategory discrimination.

Compared to the above methods, our TK-Net achieves the best overall performance with OA of 71.25%, AA of

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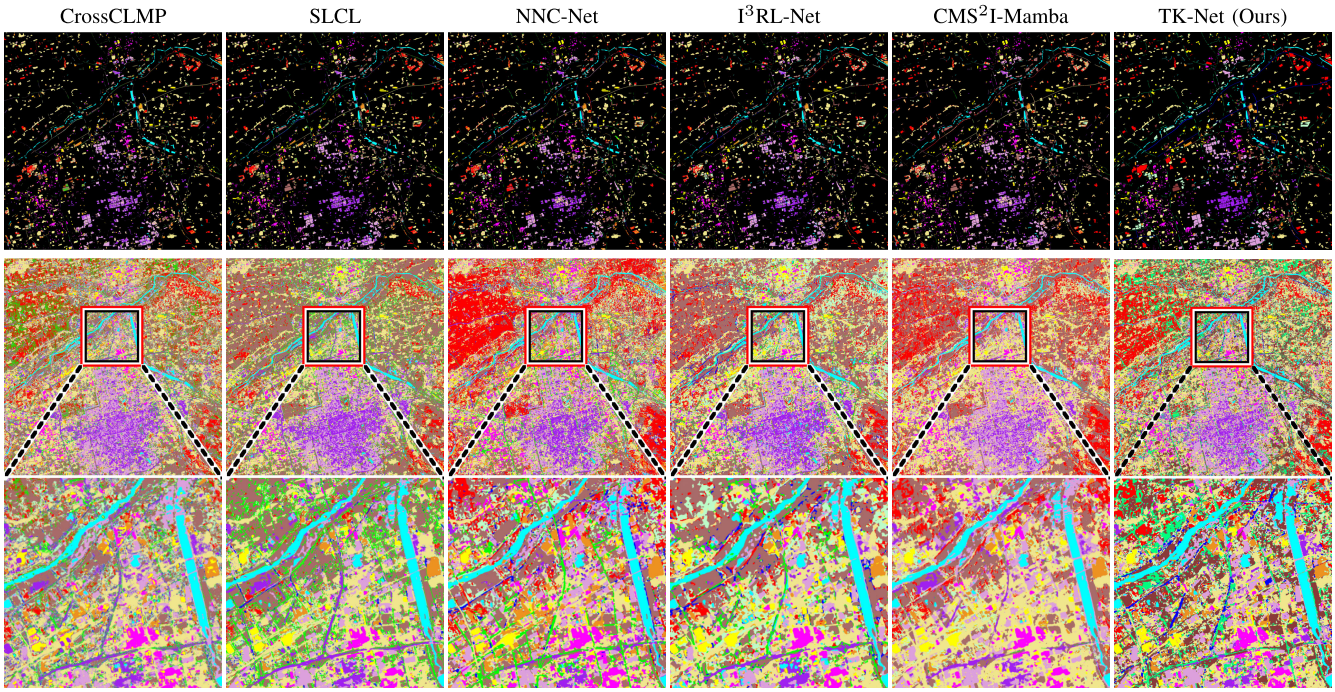



Fig. 9. Classification maps and local comparisons on the Xi'an dataset. The columns show CrossCLMP, SLCL, NNC-Net, I$^3$RL-Net, CMS$^2$I-Mamba, and TK-Net, respectively. (Top to bottom) Rows show predictions at labeled test locations and full-scene maps paired with enlarged local views; dashed lines connect the selected regions to their enlargements.

71.23%, and Kappa of 67.31%. It outperforms the second-best method NNCNet by 4.63% in OA. The improvement is particularly pronounced on challenging building categories: intensive building (c5, 74.76% versus 62.46%), sparse building (c6, 62.86% versus 70.74%), and suburb building (c7, 77.14%). We attribute this to the TAS strategy, which constructs more diverse and informative support sets that cover extreme intraclass variations, and the KMA module, which selectively aligns cross-modal features on structurally salient regions rather than indiscriminately on all spatial locations. For vegetation (c4, 81.18%) and farmland (c8, 85.83%), TK-Net also achieves competitive accuracy, benefiting from the structure-aware feature decomposition that highlights textural boundaries. For certain categories, such as red building (c10, 51.62%) and blue building (c11, 57.15%), the accuracy remains relatively low due to their severe intraclass spectral similarity and limited training samples, which is a common challenge across all methods.

The enlarged urban view in Fig. 7 qualitatively shows TK-Net retaining fine building patches and narrow road boundaries more clearly than the smoother I$^3$RL-Net result; this visual comparison complements the quantitative results in Table II.

### *F. Experimental Result on Nanjing and Xi'an Datasets*

We further evaluate the methods on the Nanjing and Xi'an datasets. The quantitative results are presented in Tables III and IV, respectively.

On the Nanjing dataset, our TK-Net maintains the highest performance with OA of 79.11%, AA of 70.21%, and Kappa of 75.42%. NNCNet again ranks second with OA of 75.00%, followed by SLCL (71.17%) and CrossCLMP (70.53%). Notably, TK-Net achieves the highest accuracy on road (c2, 69.71%), significantly exceeding the second-best NNCNet (58.98%), owing to the KMA module's ability to capture elongated linear structures through cross-modal attention. For intensive building (c5, 83.35%) and sparse building (c6, 99.39%), TK-Net also attains top performance, benefiting from the TAS strategy that ensures the support set covers the full spectrum of building densities. CMS$^2$I-Mamba performs well on sparse building (c6, 97.05%) due to its long-range spatial modeling. Still, its performance degrades on categories with subtle spectral differences, such as red building (c10, 1.20%) and white building (c9, 3.04%), where the

TABLE III
QUANTITATIVE EVALUATION RESULTS ON THE NANJING DATASET

| Methods | Cross CLMP | SLCL | NNC-Net | I$^3$RL-Net | CMS$^2$I-Mamba | TK-Net (Ours) |
|---|---|---|---|---|---|---|
| $c_1$ | 96.05 | 93.57 | 94.16 | **99.25** | 95.43 | 94.54 |
| $c_2$ | 31.53 | 43.47 | 58.98 | 32.72 | 25.13 | **69.71** |
| $c_3$ | **93.33** | 85.81 | **94.00** | 90.32 | 88.32 | 89.54 |
| $c_4$ | 83.05 | 79.84 | 80.25 | 72.97 | 84.53 | **85.21** |
| $c_5$ | 62.24 | 66.40 | 78.76 | 79.09 | 41.45 | **83.35** |
| $c_6$ | 91.38 | 82.12 | 93.08 | 83.38 | 97.05 | **99.39** |
| $c_7$ | 78.19 | 76.02 | 70.18 | 64.17 | **84.29** | 80.75 |
| $c_8$ | 52.14 | 60.93 | 85.89 | 86.96 | **90.14** | 44.62 |
| $c_9$ | 7.09 | 26.00 | 21.11 | 22.30 | 3.04 | **39.57** |
| $c_{10}$ | 17.17 | 18.19 | **35.06** | 8.61 | 1.20 | 25.48 |
| $c_{11}$ | 58.23 | 60.17 | 55.53 | 44.98 | **60.65** | 60.18 |
| OA (%) | 70.53 | 71.17 | 75.00 | 68.72 | 69.90 | **79.11** |
| AA (%) | 60.95 | 62.96 | 69.73 | 62.25 | 61.02 | **70.21** |
| Kappa (%) | 65.19 | 66.18 | 70.68 | 63.52 | 63.89 | **75.42** |

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TABLE IV
QUANTITATIVE EVALUATION RESULTS ON THE XI'AN DATASET

| Methods | Cross CLMP | SLCL | NNC-Net | $I^3$RL-Net | CMS$^2$I-Mamba | TK-Net (Ours) |
|---|---|---|---|---|---|---|
| $c_1$ | 95.69 | **98.01** | 89.62 | 97.10 | 94.18 | 96.82 |
| $c_2$ | 10.34 | **35.70** | 27.45 | 12.06 | 1.70 | 34.17 |
| $c_3$ | 37.01 | 47.47 | 33.41 | 25.05 | 11.30 | **68.96** |
| $c_4$ | 65.12 | **69.77** | 66.32 | 68.34 | 64.33 | 66.21 |
| $c_5$ | 26.81 | 28.08 | 40.86 | 24.04 | 33.18 | **44.59** |
| $c_6$ | 65.41 | 45.24 | 67.31 | **81.89** | 59.81 | 73.66 |
| $c_7$ | 89.98 | 94.44 | **94.76** | 81.92 | 93.52 | 90.47 |
| $c_8$ | 67.88 | **82.04** | 69.09 | 82.01 | 60.93 | 78.93 |
| $c_9$ | 69.41 | 66.32 | 77.49 | 68.14 | **78.90** | 74.81 |
| $c_{10}$ | 95.76 | 95.31 | **98.16** | 90.12 | 91.85 | 97.97 |
| $c_{11}$ | 92.84 | 82.76 | **95.49** | 80.64 | 92.56 | 89.34 |
| $c_{12}$ | 70.06 | 80.78 | **82.67** | 74.39 | 75.81 | 73.52 |
| OA (%) | 73.98 | 74.14 | 78.30 | 73.58 | 72.94 | **80.14** |
| AA (%) | 65.52 | 68.83 | 70.22 | 65.47 | 63.17 | **74.12** |
| Kappa (%) | 70.02 | 70.66 | 75.14 | 69.89 | 68.78 | **77.27** |

absence of explicit contrastive alignment limits its discriminative capacity.

On the Xi'an dataset, the largest and most challenging among the three, TK-Net achieves OA of 80.14%, AA of 74.12%, and Kappa of 77.27%, again securing the top position. NNCNet obtains OA of 78.30%, with SLCL at 74.14% and CrossCLMP at 73.98%. The advantages of TK-Net are especially evident in classes with high intraclass variance. For bareland (c3), TK-Net achieves 68.96%, substantially outperforming CrossCLMP (37.01%) and CMS$^2$I-Mamba (11.30%), as the TAS strategy explicitly selects dispersed support samples that cover the diverse spectral appearances of bare soil. For road (c2), TK-Net obtains 34.17%, markedly higher than CMS$^2$I-Mamba (1.70%) and CrossCLMP (10.34%). This indicates that on large-scale datasets with heterogeneous road surfaces, the feature-driven sampling of TK-Net provides more representative prototypes than random or global contrastive sampling.

Across all three datasets, TK-Net consistently achieves the highest OA, demonstrating the generalization capability of its two core components: the TAS strategy ensures diverse few-shot task construction across varying class distributions, and the KMA module enables region-selective cross-modal alignment that suppresses background interference while amplifying discriminative structural features.

### G. Computational Complexity and Inference Efficiency Analysis

Table V compares the computational complexity and inference efficiency under the Xi'an setting. TK-Net achieves 2.937 ms inference time per sample, which is faster than CrossCLMP, SLCL, $I^3$RL-Net, and CMS$^2$I-Mamba, although it is slightly slower than NNC-Net.

The computational structure is shaped by three design choices: dual-encoder architecture for cross-modal learning, feature-driven sampling in TAS, and region-selective alignment in KMA. The region-level modeling—superpixel generation, cross-modal similarity computation, key region mining, and bidirectional spatial attention—contributes 2.508G FLOPs. Compared to NNC-Net (0.123G), TK-Net incurs higher complexity, reflecting a tradeoff between region-aware representation learning and computational efficiency.

TABLE V
COMPUTATIONAL COMPLEXITY AND INFERENCE EFFICIENCY UNDER THE XI'AN EXPERIMENTAL SETTING

| Methods | Cross CLMP | SLCL | NNC-Net | $I^3$RL-Net | CMS$^2$I-Mamba | TK-Net (Ours) |
|---|---|---|---|---|---|---|
| Params (M) | 51.528 | 22.700 | 2.747 | 25.163 | 6.735 | 9.801 |
| FLOPs (G) | 5.545 | 2.362 | 0.123 | 7.259 | 0.318 | 2.508 |
| Inference Time (ms/sample) | 7.950 | 3.528 | 2.664 | 8.344 | 9.406 | 2.937 |

We acknowledge that the current complexity is not optimal. The dual-branch design increases the parameter count, and SLIC-based region processing limits parallelization. Future work will explore learnable region grouping, feature caching, and selective pruning to reduce overhead while maintaining region-selective alignment.

## V. CONCLUSION

In this article, we propose TK-Net, a novel few-shot cross-modal classification framework for remote sensing images. Experiments on three datasets verified its effectiveness in fully exploiting informative samples and fine-grained cross-modal features, thereby contributing to classification. TK-Net explicitly tackles the bottlenecks of few-shot cross-modal tasks through two core modules. However, this region-aware and dual-branch optimization method requires generating superpixels and processing multiple contrastive pairs separately, resulting in a slightly higher overall parameter count and computational overhead. In future work, we will explore more efficient implementations of TK-Net, such as lightweight superpixel approximation, reusable region-level feature caches, and simplified contrastive pair construction, to reduce computational overhead while maintaining competitive few-shot classification performance.

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